

Failure of GCH on a measurable with the Ultrapower Axiom

Eyal Kaplan
Joint Work with Omer Ben-Neria

September 2024

- 1 The Ultrapower Axiom
- 2 The Friedman-Magidor blueprint
- 3 The Kunen-like blueprint

The Ultrapower Axiom

The Ultrapower Axiom (UA)

Let U, W be a pair of uniform, σ -complete ultrafilters. Then there exist uniform, σ -complete ultrafilters $U^* \in M_W, W^* \in M_U$ such that $M_{U^*}^{M_W} = M_{W^*}^{M_U}$ and $j_{U^*}^{M_W} \circ j_W = j_{W^*}^{M_U} \circ j_U$.

The Ultrapower Axiom

The Ultrapower Axiom (UA)

Let U, W be a pair of uniform, σ -complete ultrafilters. Then there exist uniform, σ -complete ultrafilters $U^* \in M_W, W^* \in M_U$ such that $M_{U^*}^{M_W} = M_{W^*}^{M_U}$ and $j_{U^*}^{M_U} \circ j_W = j_{W^*}^{M_W} \circ j_U$.

$$\begin{array}{ccc} & \text{Ult}(M_U, W^*) \simeq \text{Ult}(M_W, U^*) & \\ j_{W^*}^{M_U} \nearrow & & \nwarrow j_{U^*}^{M_W} \\ W^* \in M_U & & M_W \ni U^* \\ j_U \nwarrow & & \nearrow j_W \\ & V & \end{array}$$

Consequences of UA

(Goldberg) The following are consequences of the Ultrapower Axiom:

- 1 The Mitchell order is linear.

Consequences of UA

(Goldberg) The following are consequences of the Ultrapower Axiom:

- 1 The Mitchell order is linear.
- 2 Assume that κ is the least measurable cardinal, and let U be the unique normal measure on κ . Then every nontrivial, σ -complete ultrafilter on κ is isomorphic to U^n for some $n < \omega$.

Consequences of UA

(Goldberg) The following are consequences of the Ultrapower Axiom:

- 1 The Mitchell order is linear.
- 2 Assume that κ is the least measurable cardinal, and let U be the unique normal measure on κ . Then every nontrivial, σ -complete ultrafilter on κ is isomorphic to U^n for some $n < \omega$.
- 3 The least Strongly Compact cardinal is Supercompact.

Consequences of UA

(Goldberg) The following are consequences of the Ultrapower Axiom:

- 1 The Mitchell order is linear.
- 2 Assume that κ is the least measurable cardinal, and let U be the unique normal measure on κ . Then every nontrivial, σ -complete ultrafilter on κ is isomorphic to U^n for some $n < \omega$.
- 3 The least Strongly Compact cardinal is Supercompact. Moreover, every Strongly Compact cardinal is either Supercompact, or a measurable limit of Supercompacts.

(Goldberg) The following are consequences of the Ultrapower Axiom:

- 1 The Mitchell order is linear.
- 2 Assume that κ is the least measurable cardinal, and let U be the unique normal measure on κ . Then every nontrivial, σ -complete ultrafilter on κ is isomorphic to U^n for some $n < \omega$.
- 3 The least Strongly Compact cardinal is Supercompact. Moreover, every Strongly Compact cardinal is either Supercompact, or a measurable limit of Supercompacts.
- 4 Assume that κ is Supercompact. Then $\text{GCH}_{\geq \kappa}$ holds.

UA + Failure of GCH on a Measurable

Question (Goldberg)

Is UA consistent with the failure of GCH on a measurable cardinal κ ?

Question (Goldberg)

Is UA consistent with the failure of GCH on a measurable cardinal κ ?

- Starting from an inner model with suitably large cardinals, one can achieve the failure of GCH on a measurable, by performing an iterated forcing of length $\kappa + 1$.

Question (Goldberg)

Is UA consistent with the failure of GCH on a measurable cardinal κ ?

- Starting from an inner model with suitably large cardinals, one can achieve the failure of GCH on a measurable, by performing an iterated forcing of length $\kappa + 1$.
- However, many known methods to do so fail to preserve the UA, since they violate the linearity of the Mitchell order.

Question (Goldberg)

Is UA consistent with the failure of GCH on a measurable cardinal κ ?

- Starting from an inner model with suitably large cardinals, one can achieve the failure of GCH on a measurable, by performing an iterated forcing of length $\kappa + 1$.
- However, many known methods to do so fail to preserve the UA, since they violate the linearity of the Mitchell order.
- Other methods preserve the linearity of the Mitchell order, but not the Ultrapower Axiom.

- **Friedman - Magidor, 2009:** Starting from a cardinal κ which is $(\kappa + 2)$ -strong, forced the failure of *GCH* at the measurable κ , with a unique normal measure on it.

- **Friedman - Magidor, 2009:** Starting from a cardinal κ which is $(\kappa + 2)$ -strong, forced the failure of *GCH* at the measurable κ , with a unique normal measure on it.
- **Ben-Neria - Gitik, 2017:** Starting from a cardinal κ with $o(\kappa) = \kappa^{++}$, forced the failure of *GCH* at κ , with a unique normal measure on it.

- **Friedman - Magidor, 2009:** Starting from a cardinal κ which is $(\kappa + 2)$ -strong, forced the failure of *GCH* at the measurable κ , with a unique normal measure on it.
- **Ben-Neria - Gitik, 2017:** Starting from a cardinal κ with $o(\kappa) = \kappa^{++}$, forced the failure of *GCH* at κ , with a unique normal measure on it.
- **Apter - Cummings, 2023:** Starting from a strong cardinal κ , forced the failure of *GCH* on a measurable cardinal, preserving the strongness of κ and the linearity of the Mitchell order on it.

- **Friedman - Magidor, 2009:** Starting from a cardinal κ which is $(\kappa + 2)$ -strong, forced the failure of *GCH* at the measurable κ , with a unique normal measure on it.
- **Ben-Neria - Gitik, 2017:** Starting from a cardinal κ with $o(\kappa) = \kappa^{++}$, forced the failure of *GCH* at κ , with a unique normal measure on it.
- **Apter - Cummings, 2023:** Starting from a strong cardinal κ , forced the failure of *GCH* on a measurable cardinal, preserving the strongness of κ and the linearity of the Mitchell order on it. (since κ remains strong in the generic extension, it carries $(2^\kappa)^+$ -many normal measures).

The Main Result

Theorem (Ben-Neria, K.)

Let $L[\vec{E}]$ be the minimal inner model for the existence of a measurable cardinal κ carrying a (κ, κ^{++}) -extender. Then there exists a forcing extension in which GCH fails at κ and UA holds.

The Main Result

Theorem (Ben-Neria, K.)

Let $L[\vec{E}]$ be the minimal inner model for the existence of a measurable cardinal κ carrying a (κ, κ^{++}) -extender. Then there exists a forcing extension in which GCH fails at κ and UA holds.

- The forcing is a variation of the Friedman-Magidor forcing.

The Main Result

Theorem (Ben-Neria, K.)

Let $L[\vec{E}]$ be the minimal inner model for the existence of a measurable cardinal κ carrying a (κ, κ^{++}) -extender. Then there exists a forcing extension in which *GCH* fails at κ and *UA* holds.

- The forcing is a variation of the Friedman-Magidor forcing.
- We present below two blueprints:

The Main Result

Theorem (Ben-Neria, K.)

Let $L[\vec{E}]$ be the minimal inner model for the existence of a measurable cardinal κ carrying a (κ, κ^{++}) -extender. Then there exists a forcing extension in which GCH fails at κ and UA holds.

- The forcing is a variation of the Friedman-Magidor forcing.
- We present below two blueprints:
 - The first, **the Friedman-Magidor blueprint**, extracts the properties of the Friedman Magidor forcing, that ensure that GCH breaks at a measurable cardinal κ carrying a unique normal measure U .

The Main Result

Theorem (Ben-Neria, K.)

Let $L[\vec{E}]$ be the minimal inner model for the existence of a measurable cardinal κ carrying a (κ, κ^{++}) -extender. Then there exists a forcing extension in which GCH fails at κ and UA holds.

- The forcing is a variation of the Friedman-Magidor forcing.
- We present below two blueprints:
 - The first, **the Friedman-Magidor blueprint**, extracts the properties of the Friedman Magidor forcing, that ensure that GCH breaks at a measurable cardinal κ carrying a unique normal measure U .
 - The second, **the Kunen-like blueprint**, extends the Friedman-Magidor blueprint by adding two additional requirements, ensuring that every σ -complete ultrafilter is Rudin-Keisler equivalent to U^n , for some $n < \omega$.

The Main Result

Theorem (Ben-Neria, K.)

Let $L[\vec{E}]$ be the minimal inner model for the existence of a measurable cardinal κ carrying a (κ, κ^{++}) -extender. Then there exists a forcing extension in which GCH fails at κ and UA holds.

- The forcing is a variation of the Friedman-Magidor forcing.
- We present below two blueprints:
 - The first, **the Friedman-Magidor blueprint**, extracts the properties of the Friedman Magidor forcing, that ensure that GCH breaks at a measurable cardinal κ carrying a unique normal measure U .
 - The second, **the Kunen-like blueprint**, extends the Friedman-Magidor blueprint by adding two additional requirements, ensuring that every σ -complete ultrafilter is Rudin-Keisler equivalent to U^n , for some $n < \omega$.
- An important part of the work (omitted here) is the construction of a forcing notion that fulfills the Kunen-like blueprint. One of the main features of the construction is the integration of fine-structure into it.

The Friedman-Magidor blueprint

Work in $L[\bar{E}]$. Assume that $\langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha : \alpha \leq \kappa + 1 \rangle$ is an iterated forcing. We say that it fulfils the Friedman-Magidor blueprint if it satisfies the following properties:

The Friedman-Magidor blueprint

- ① For each $\alpha \leq \kappa$, the $\mathbb{P}_\alpha$ -name $\dot{\mathbb{Q}}_\alpha$ is the trivial poset unless α is Mahlo.
- ② For every Mahlo cardinal $\alpha \leq \kappa$, $0_{\mathbb{P}_\alpha}$ forces $\dot{\mathbb{Q}}_\alpha$ satisfies the following requirements:
 - $\dot{\mathbb{Q}}_\alpha \in V_\kappa$ (unless $\alpha = \kappa$), has size $|\dot{\mathbb{Q}}_\alpha| = \alpha^{++}$ and adds α^{++} new subsets to α ,
 - $\dot{\mathbb{Q}}_\alpha$ is α -distributive, satisfies the α^{++} .c.c, and preserves the Mahloness of α .
 - $\dot{\mathbb{Q}}_\alpha$ satisfies the α Continuous-Fusion property*,
 - $\dot{\mathbb{Q}}_\alpha$ is self coding**.

*Fusion arguments played an important role in the Friedman-Magidor forcing. The " α Continuous-Fusion property" and the "Iteration-Fusion property" (which will be mentioned below), are abstract properties allowing to perform those arguments.

**The notion of "Self-Coding forcing" will be properly defined below.

The Friedman-Magidor blueprint - continued

- ③ For every regular cardinal $\alpha \leq \kappa$,
 - $\mathbb{P}_\alpha$ has the α Iteration-Fusion property,
 - $0_{\mathbb{P}_\alpha} \Vdash (\mathbb{P}/\mathbb{P}_\alpha)$ is α -distributive.
- ④ The iteration $\mathbb{P} = \mathbb{P}_{\kappa+1}$ is uniformly definable in the sense that there is a parameter-free formula $\varphi(x)$ such that, for every $\alpha \leq \kappa$, $\varphi(\alpha)$ defines $\mathbb{P}_{\alpha+1}$ and some relevant parameters*** correctly over every inner Model that contains $H_{\alpha^{++}}$.

***Other than $\mathbb{P}_{\alpha+1}$ itself, the formula should also define some parameters related to Self-Coding, and the $\mathbb{P}_{\alpha+1}$ -names for the α^{++} -many subsets added to α by $\mathbb{Q}_\alpha$.

The Friedman-Magidor blueprint - continued

- ③ For every regular cardinal $\alpha \leq \kappa$,
 - $\mathbb{P}_\alpha$ has the α Iteration-Fusion property,
 - $0_{\mathbb{P}_\alpha} \Vdash (\mathbb{P}/\mathbb{P}_\alpha)$ is α -distributive.
- ④ The iteration $\mathbb{P} = \mathbb{P}_{\kappa+1}$ is uniformly definable in the sense that there is a parameter-free formula $\varphi(x)$ such that, for every $\alpha \leq \kappa$, $\varphi(\alpha)$ defines $\mathbb{P}_{\alpha+1}$ and some relevant parameters*** correctly over every inner Model that contains $H_{\alpha^{++}}$.

***Other than $\mathbb{P}_{\alpha+1}$ itself, the formula should also define some parameters related to Self-Coding, and the $\mathbb{P}_{\alpha+1}$ -names for the α^{++} -many subsets added to α by $\mathbb{Q}_\alpha$.

The Friedman-Magidor blueprint

Theorem (Friedman-Magidor)

Assume $L[\bar{E}]$ is the minimal inner model with a measurable cardinal κ carrying a (κ, κ^{++}) extender, and let $\mathbb{P} = \mathbb{P}_{\kappa+1}$ be an iteration satisfying the Friedman-Magidor blueprint. Then $\mathbb{P}$ forces that κ is the least measurable, $2^\kappa = \kappa^{++}$, and there exists a unique normal measure on κ .

Main ideas in the proof that κ carries a unique normal measure

- Assume that $G \subseteq \mathbb{P}_{\kappa+1}$ is $\mathbb{P}_{\kappa+1}$ -generic over $V = L[\bar{E}]$, and $U \in V[G]$ is a normal measure on κ .

Main ideas in the proof that κ carries a unique normal measure

- Assume that $G \subseteq \mathbb{P}_{\kappa+1}$ is $\mathbb{P}_{\kappa+1}$ -generic over $V = L[\bar{E}]$, and $U \in V[G]$ is a normal measure on κ .
- Let $j_U: V[G] \rightarrow M[H]$ be the associated ultrapower embedding.

Main ideas in the proof that κ carries a unique normal measure

- Assume that $G \subseteq \mathbb{P}_{\kappa+1}$ is $\mathbb{P}_{\kappa+1}$ -generic over $V = L[\bar{E}]$, and $U \in V[G]$ is a normal measure on κ .
- Let $j_U: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. It's enough to show the following:

Main ideas in the proof that κ carries a unique normal measure

- Assume that $G \subseteq \mathbb{P}_{\kappa+1}$ is $\mathbb{P}_{\kappa+1}$ -generic over $V = L[\bar{E}]$, and $U \in V[G]$ is a normal measure on κ .
- Let $j_U: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. It's enough to show the following:

Properties ensuring that U is uniquely determined:

- 1 $M = M_E$, $j_E \subseteq j_U$, where E is the top (κ, κ^{++}) -extender in $L[\bar{E}]$.

Main ideas in the proof that κ carries a unique normal measure

- Assume that $G \subseteq \mathbb{P}_{\kappa+1}$ is $\mathbb{P}_{\kappa+1}$ -generic over $V = L[\bar{E}]$, and $U \in V[G]$ is a normal measure on κ .
- Let $j_U: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. It's enough to show the following:

Properties ensuring that U is uniquely determined:

- 1 $M = M_E$, $j_E \subseteq j_U$, where E is the top (κ, κ^{++}) -extender in $L[\bar{E}]$.
- 2 H is unique, in the sense that it is the only $j_E(\mathbb{P}_{\kappa+1})$ -generic set over M in $V[G]$, satisfying $j_E[G] \subseteq H$.

Main ideas in the proof that κ carries a unique normal measure

- Assume that $G \subseteq \mathbb{P}_{\kappa+1}$ is $\mathbb{P}_{\kappa+1}$ -generic over $V = L[\bar{E}]$, and $U \in V[G]$ is a normal measure on κ .
- Let $j_U: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. It's enough to show the following:

Properties ensuring that U is uniquely determined:

- 1 $M = M_E$, $j_E \subseteq j_U$, where E is the top (κ, κ^{++}) -extender in $L[\bar{E}]$.
- 2 H is unique, in the sense that it is the only $j_E(\mathbb{P}_{\kappa+1})$ -generic set over M in $V[G]$, satisfying $j_E[G] \subseteq H$.

$$① \quad M = M_E, j_E \subseteq j_U$$

The proof that $j_U \restriction V = j_E$ and $M = M_E$ goes as follows:

- $j_U \restriction V$ is an iterated ultrapower by extenders. Letting E_0 being the first extender applied in the normal iteration, we first show that $E_0 = E$ is the top (κ, κ^{++}) -extender.
- Indeed, otherwise,

$$\kappa < j_{E_0}(\kappa) \leq \kappa^{++} \leq (2^\kappa)^{M[H]}$$

contradicting the fact that $j_{E_0}(\kappa)$ is strongly inaccessible in $M[H]$.

$$① \quad M = M_E, j_E \subseteq j_U$$

The proof that $j_U \restriction V = j_E$ and $M = M_E$ goes as follows:

- $j_U \restriction V$ is an iterated ultrapower by extenders. Letting E_0 being the first extender applied in the normal iteration, we first show that $E_0 = E$ is the top (κ, κ^{++}) -extender.
- Indeed, otherwise,

$$\kappa < j_{E_0}(\kappa) \leq \kappa^{++} \leq (2^\kappa)^{M[H]}$$

contradicting the fact that $j_{E_0}(\kappa)$ is strongly inaccessible in $M[H]$.

- The Fusion properties of $\mathbb{P}_\kappa, \mathbb{Q}_\kappa$ can be used to approximate functions in $(\text{Ord}^\kappa)^{V[G]}$ by ground model functions; this fact can be used to deduce that $M = M_E$ and $j_U \restriction V = j_E$.

② H is unique

The next step is to prove the uniqueness of $H \in V[G]$ which is $j_E(\mathbb{P})$ -generic over M_E , with $j_E[G] \subseteq H$. Very roughly, this is done as follows:

② H is unique

The next step is to prove the uniqueness of $H \in V[G]$ which is $j_E(\mathbb{P})$ -generic over M_E , with $j_E[G] \subseteq H$. Very roughly, this is done as follows:

- By standard arguments, $H \restriction \kappa = G \restriction \kappa$.

② H is unique

The next step is to prove the uniqueness of $H \in V[G]$ which is $j_E(\mathbb{P})$ -generic over M_E , with $j_E[G] \subseteq H$. Very roughly, this is done as follows:

- By standard arguments, $H \restriction \kappa = G \restriction \kappa$.
- The fact that each $\dot{\mathbb{Q}}_\alpha$ is self-coding implies that $H(\kappa) = G(\kappa)$.

2 H is unique

The next step is to prove the uniqueness of $H \in V[G]$ which is $j_E(\mathbb{P})$ -generic over M_E , with $j_E[G] \subseteq H$. Very roughly, this is done as follows:

- By standard arguments, $H \restriction \kappa = G \restriction \kappa$.
- The fact that each $\dot{\mathbb{Q}}_\alpha$ is self-coding implies that $H(\kappa) = G(\kappa)$.
- The fact that $\mathbb{P}_\kappa$ satisfies the κ Iteration-Fusion property implies that $H(\kappa, j_E(\kappa))$ is generated from $j_E[G] \setminus \kappa + 1 = \{j_E(p) \setminus \kappa + 1 : p \in G \restriction \kappa\}$.

2 H is unique

The next step is to prove the uniqueness of $H \in V[G]$ which is $j_E(\mathbb{P})$ -generic over M_E , with $j_E[G] \subseteq H$. Very roughly, this is done as follows:

- By standard arguments, $H \restriction \kappa = G \restriction \kappa$.
- The fact that each $\dot{\mathbb{Q}}_\alpha$ is self-coding implies that $H(\kappa) = G(\kappa)$.
- The fact that $\mathbb{P}_\kappa$ satisfies the κ Iteration-Fusion property implies that $H(\kappa, j_E(\kappa))$ is generated from $j_E[G] \setminus \kappa + 1 = \{j_E(p) \setminus \kappa + 1 : p \in G \restriction \kappa\}$.
- The fact that each $\mathbb{Q}_\alpha$ satisfies the α C-Fusion property implies that $H(j_E(\kappa))$ is the unique $j_E(\dot{\mathbb{Q}}_\kappa)$ -generic in $V[G]$ that contains $\bar{j}[G(\kappa)]$ (here, $\bar{j}: V[G \restriction \kappa] \rightarrow M_E[j_E[G \restriction \kappa]]$ denotes the natural lift of j_E).

2 H is unique

The next step is to prove the uniqueness of $H \in V[G]$ which is $j_E(\mathbb{P})$ -generic over M_E , with $j_E[G] \subseteq H$. Very roughly, this is done as follows:

- By standard arguments, $H \restriction \kappa = G \restriction \kappa$.
- The fact that each $\dot{\mathbb{Q}}_\alpha$ is self-coding implies that $H(\kappa) = G(\kappa)$.
- The fact that $\mathbb{P}_\kappa$ satisfies the κ Iteration-Fusion property implies that $H(\kappa, j_E(\kappa))$ is generated from $j_E[G] \setminus \kappa + 1 = \{j_E(p) \setminus \kappa + 1 : p \in G \restriction \kappa\}$.
- The fact that each $\mathbb{Q}_\alpha$ satisfies the α C-Fusion property implies that $H(j_E(\kappa))$ is the unique $j_E(\dot{\mathbb{Q}}_\kappa)$ -generic in $V[G]$ that contains $\bar{j}[G(\kappa)]$ (here, $\bar{j}: V[G \restriction \kappa] \rightarrow M_E[j_E[G \restriction \kappa]]$ denotes the natural lift of j_E).

Below, we will elaborate on the definition of Self Coding and the proof that it implies that $H(\kappa) = G(\kappa)$.

Definition - Self-Coding

Let $\mathbb{Q}$ be a poset, $\vec{q} = \langle q_\alpha : \alpha < |\mathbb{Q}| \rangle$ be an enumeration of $\mathbb{Q}$ and $\vec{S} = \langle S_\alpha : \alpha < |\mathbb{Q}| \rangle$ be a sequence of pairwise almost-disjoint stationary subsets of λ for some regular $\lambda \leq |\mathbb{Q}|$.

Definition - Self-Coding

Let $\mathbb{Q}$ be a poset, $\vec{q} = \langle q_\alpha : \alpha < |\mathbb{Q}| \rangle$ be an enumeration of $\mathbb{Q}$ and $\vec{S} = \langle S_\alpha : \alpha < |\mathbb{Q}| \rangle$ be a sequence of pairwise almost-disjoint stationary subsets of λ for some regular $\lambda \leq |\mathbb{Q}|$.

We say that $\mathbb{Q}$ **admits self coding** via $\vec{q}, \vec{S}$ if and only if, for every $\tau < |\mathbb{Q}|$,

Definition - Self-Coding

Let $\mathbb{Q}$ be a poset, $\vec{q} = \langle q_\alpha : \alpha < |\mathbb{Q}| \rangle$ be an enumeration of $\mathbb{Q}$ and $\vec{S} = \langle S_\alpha : \alpha < |\mathbb{Q}| \rangle$ be a sequence of pairwise almost-disjoint stationary subsets of λ for some regular $\lambda \leq |\mathbb{Q}|$.

We say that $\mathbb{Q}$ **admits self coding** via $\vec{q}, \vec{S}$ if and only if, for every $\tau < |\mathbb{Q}|$,

$$0_{\mathbb{Q}} \Vdash (\check{S}_{2\tau} \text{ is not stationary} \iff q_\tau \in \dot{G}_{\mathbb{Q}}) \text{ and}$$

$$0_{\mathbb{Q}} \Vdash (\check{S}_{2\tau+1} \text{ is not stationary} \iff q_\tau \notin \dot{G}_{\mathbb{Q}}).$$

where $\dot{G}_{\mathbb{Q}}$ denotes the canonical $\mathbb{Q}$ -name for the generic set.

Definition - Self-Coding

Let $\mathbb{Q}$ be a poset, $\vec{q} = \langle q_\alpha : \alpha < |\mathbb{Q}| \rangle$ be an enumeration of $\mathbb{Q}$ and $\vec{S} = \langle S_\alpha : \alpha < |\mathbb{Q}| \rangle$ be a sequence of pairwise almost-disjoint stationary subsets of λ for some regular $\lambda \leq |\mathbb{Q}|$.

We say that $\mathbb{Q}$ **admits self coding** via $\vec{q}, \vec{S}$ if and only if, for every $\tau < |\mathbb{Q}|$,

$$0_{\mathbb{Q}} \Vdash (\check{S}_{2\tau} \text{ is not stationary} \iff q_\tau \in \dot{G}_{\mathbb{Q}}) \text{ and}$$

$$0_{\mathbb{Q}} \Vdash (\check{S}_{2\tau+1} \text{ is not stationary} \iff q_\tau \notin \dot{G}_{\mathbb{Q}}).$$

where $\dot{G}_{\mathbb{Q}}$ denotes the canonical $\mathbb{Q}$ -name for the generic set.

In the blueprints, we require that each iterate $\mathbb{Q}_\alpha$ ($\alpha \leq \kappa$) satisfies Self-Coding. We also require that the lists $\vec{q}, \vec{S}$ witnessing that, have uniformly definable $\mathbb{P}_\alpha$ -names over $H_{\alpha^{++}}$.

The uniqueness of $H(\kappa)$

Using the self coding, we can prove that $H(\kappa) = G(\kappa)$.

- Assume that the self-coding of $\mathbb{Q}_\kappa$ is witnessed by sequences $\langle q_\alpha : \alpha < \kappa^{++} \rangle, \langle S_\alpha : \alpha < \kappa^{++} \rangle$.
- The same sequences witness the self coding of $\mathbb{Q}_\kappa$ over $M_E[G_{<\kappa}]$.
- Therefore, for every $\tau < \kappa^{++}$,

$$q_\tau \in G(\kappa) \iff V[G] \models S_{2\tau+1} \text{ is stationary} \implies \\ M_E[G] \models S_{2\tau+1} \text{ is stationary} \iff q_\tau \in H(\kappa).$$

The Kunen-Like blueprint

The Kunen-Like blueprint

The Kunen-Like blueprint is obtained from the Friedman-Magidor blueprint by adding the following assumptions:

- For every Mahlo cardinal $\alpha \leq \kappa$, $\dot{\mathbb{Q}}_\alpha$ adds an increasing sequence in $<_{Sing}^\alpha$ of length α^{++} .
(for $f, g \in \alpha^\alpha$, we say that $f <_{Sing}^\alpha g$ if there exists a club $C \subseteq \alpha$ such that for every singular $i \in C$, $f(i) < g(i)$).
- The self coding of $\dot{\mathbb{Q}}_\alpha$ is being done by sequences $\vec{q}^\alpha$ and $\vec{S}^\alpha$, where $\vec{S}^\alpha = \langle S_\tau^\alpha \mid \tau < \alpha^{++} \rangle$ consists of almost disjoint stationary subsets of $\alpha^+ \cap \text{Cof}(\rho_\alpha)$, for some regular cardinal $\rho_\alpha < \alpha$.

The main Theorem - proof sketch

- 1 Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.

The main Theorem - proof sketch

- 1 Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.
- 2 In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

The main Theorem - proof sketch

- 1 Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.
- 2 In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

The main Theorem - proof sketch

① Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.

② In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

③ Identifying n : Let $j_W: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. Then $j_W \restriction V$ is an iterated ultrapower of $L[\bar{E}]$ by its extenders.

The main Theorem - proof sketch

① Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.

② In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

③ Identifying n : Let $j_W: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. Then $j_W \upharpoonright V$ is an iterated ultrapower of $L[\bar{E}]$ by its extenders. Since $\mathbb{P}$ is ω_1 -distributive, the iteration must have a finite length - denote it by n .

The main Theorem - proof sketch

- 1 Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.
- 2 In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

- 3 Identifying n : Let $j_W: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. Then $j_W \restriction V$ is an iterated ultrapower of $L[\bar{E}]$ by its extenders. Since $\mathbb{P}$ is ω_1 -distributive, the iteration must have a finite length - denote it by n .

In order to show that $W \equiv_{RK} U^n$, it's enough to prove:

The main Theorem - proof sketch

① Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.

② In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

③ Identifying n : Let $j_W: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. Then $j_W \upharpoonright V$ is an iterated ultrapower of $L[\bar{E}]$ by its extenders. Since $\mathbb{P}$ is ω_1 -distributive, the iteration must have a finite length - denote it by n .

In order to show that $W \equiv_{RK} U^n$, it's enough to prove:

① $M = \text{Ult}(V, E^n)$, where E is the top (κ, κ^{++}) -extender of $L[\bar{E}]$.

The main Theorem - proof sketch

① Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.

② In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

③ Identifying n : Let $j_W: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. Then $j_W \upharpoonright V$ is an iterated ultrapower of $L[\bar{E}]$ by its extenders. Since $\mathbb{P}$ is ω_1 -distributive, the iteration must have a finite length - denote it by n .

In order to show that $W \equiv_{RK} U^n$, it's enough to prove:

① $M = \text{Ult}(V, E^n)$, where E is the top (κ, κ^{++}) -extender of $L[\bar{E}]$.

② H is unique, and is actually equal to $j_{U^n}(G)$.

The main Theorem - proof sketch

① Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.

② In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

③ Identifying n : Let $j_W: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. Then $j_W \upharpoonright V$ is an iterated ultrapower of $L[\bar{E}]$ by its extenders. Since $\mathbb{P}$ is ω_1 -distributive, the iteration must have a finite length - denote it by n .

In order to show that $W \equiv_{RK} U^n$, it's enough to prove:

① $M = \text{Ult}(V, E^n)$, where E is the top (κ, κ^{++}) -extender of $L[\bar{E}]$.

② H is unique, and is actually equal to $j_{U^n}(G)$.

For simplicity, we concentrate on the case where $n=2$.

① The proof that $M = M_{E^2}$

- Assume that $j_W \upharpoonright V$ is a two step iteration, given in its normal form. Arguing as in the Friedman-Magidor blueprint, the first step must be an ultrapower with the (κ, κ^{++}) -extender E .

1 The proof that $M = M_{E^2}$

- Assume that $j_W \restriction V$ is a two step iteration, given in its normal form. Arguing as in the Friedman-Magidor blueprint, the first step must be an ultrapower with the (κ, κ^{++}) -extender E .
- Denote by $F \in \text{Ult}(L[\bar{E}], E)$ the second extender which is being applied.

1 The proof that $M = M_{E^2}$

- Assume that $j_W \upharpoonright V$ is a two step iteration, given in its normal form. Arguing as in the Friedman-Magidor blueprint, the first step must be an ultrapower with the (κ, κ^{++}) -extender E .
- Denote by $F \in \text{Ult}(L[\bar{E}], E)$ the second extender which is being applied.

Claim

$$F = j_E(E).$$

Claim

$$F = j_E(E).$$

Proof Sketch

Denote $\lambda = \text{crit}(j_F)$.

Claim

$$F = j_E(E).$$

Proof Sketch

Denote $\lambda = \text{crit}(j_F)$. Then $\lambda = j_E(\kappa)$ and $F = j_E(E)$, since:

$$(\lambda^{++})^{M_E} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{\text{Ult}(V[G], U)} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{M[H]} < j_F(\lambda)$$

Claim

$$F = j_E(E).$$

Proof Sketch

Denote $\lambda = \text{crit}(j_F)$. Then $\lambda = j_E(\kappa)$ and $F = j_E(E)$, since:

$$(\lambda^{++})^{M_E} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{\text{Ult}(V[G], U)} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{M[H]} < j_F(\lambda)$$

- The leftmost inequality follows since $\lambda \leq j_E(\kappa)$ and thus $j_E(\mathbb{P})$ adds an increasing $<_{\text{Sing}}^\lambda$ sequence of length $(\lambda^{++})^{M_E}$.

Claim

$$F = j_E(E).$$

Proof Sketch

Denote $\lambda = \text{crit}(j_F)$. Then $\lambda = j_E(\kappa)$ and $F = j_E(E)$, since:

$$(\lambda^{++})^{M_E} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{\text{Ult}(V[G], U)} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{M[H]} < j_F(\lambda)$$

- The leftmost inequality follows since $\lambda \leq j_E(\kappa)$ and thus $j_E(\mathbb{P})$ adds an increasing $<_{\text{Sing}}^\lambda$ sequence of length $(\lambda^{++})^{M_E}$.
- For the middle inequality:
 - Since U is the unique normal measure on κ in $V[G]$ and $\text{crit}(j_W) = \kappa$, $U = \{X \subseteq \kappa : \kappa \in j_W(X)\}$.

Claim

$$F = j_E(E).$$

Proof Sketch

Denote $\lambda = \text{crit}(j_F)$. Then $\lambda = j_E(\kappa)$ and $F = j_E(E)$, since:

$$(\lambda^{++})^{M_E} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{\text{Ult}(V[G], U)} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{M[H]} < j_F(\lambda)$$

- The leftmost inequality follows since $\lambda \leq j_E(\kappa)$ and thus $j_E(\mathbb{P})$ adds an increasing $<_{\text{Sing}}^\lambda$ sequence of length $(\lambda^{++})^{M_E}$.
- For the middle inequality:
 - Since U is the unique normal measure on κ in $V[G]$ and $\text{crit}(j_W) = \kappa$, $U = \{X \subseteq \kappa : \kappa \in j_W(X)\}$.
 - Therefore, there exists an embedding $k: \text{Ult}(V[G], U) \rightarrow M[H]$ such that $j_W = k \circ j_U$. It could be shown that $\text{crit}(k) \geq \lambda$.
 - Thus $(\mathcal{P}(\lambda))^{\text{Ult}(V[G], U)} \subseteq (\mathcal{P}(\lambda))^{M[H]}$, which implies the inequality.

Claim

$$F = j_E(E).$$

Proof Sketch

Denote $\lambda = \text{crit}(j_F)$. Then $\lambda = j_E(\kappa)$ and $F = j_E(E)$, since:

$$(\lambda^{++})^{M_E} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{\text{Ult}(V[G], U)} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{M[H]} < j_F(\lambda)$$

- The leftmost inequality follows since $\lambda \leq j_E(\kappa)$ and thus $j_E(\mathbb{P})$ adds an increasing $<_{\text{Sing}}^\lambda$ sequence of length $(\lambda^{++})^{M_E}$.
- For the middle inequality:
 - Since U is the unique normal measure on κ in $V[G]$ and $\text{crit}(j_W) = \kappa$, $U = \{X \subseteq \kappa : \kappa \in j_W(X)\}$.
 - Therefore, there exists an embedding $k: \text{Ult}(V[G], U) \rightarrow M[H]$ such that $j_W = k \circ j_U$. It could be shown that $\text{crit}(k) \geq \lambda$.
 - Thus $(\mathcal{P}(\lambda))^{\text{Ult}(V[G], U)} \subseteq (\mathcal{P}(\lambda))^{M[H]}$, which implies the inequality.
- The rightmost inequality follows since $j_F(\lambda)$ is inaccessible in M , and remains such in $M[H]$.

2 The proof that H is uniquely determined

The set $H \subseteq j_{E^2}(\mathbb{P})$ is the unique generic over M_{E^2} with $j_{E^2}[G] \subseteq H$. This is shown in the following steps:

- $H \restriction j_E(\kappa) = j_U(G \restriction \kappa)$: this follows as in the Friedman-Magidor blueprint.
- $H(j_E(\kappa)) = j_U(G)(j_E(\kappa))$: this uses the additional assumptions regarding the Self-Coding procedure in the Kunen-like blueprint (we will elaborate on this point).
- $H \restriction (j_E(\kappa), j_{E^2}(\kappa)]$ is uniquely determined: this follows by Fusion arguments.

Claim

$$H(j_E(\kappa)) = j_U(G)(j_E(\kappa)).$$

Proof Sketch

For simplicity, we denote $\kappa_1 = j_E(\kappa)$.

Claim

$$H(j_E(\kappa)) = j_U(G)(j_E(\kappa)).$$

Proof Sketch

For simplicity, we denote $\kappa_1 = j_E(\kappa)$.

- 1 It suffices to argue that $j_U[G(\kappa)] \subseteq H(\kappa_1)$.

Claim

$$H(j_E(\kappa)) = j_U(G)(j_E(\kappa)).$$

Proof Sketch

For simplicity, we denote $\kappa_1 = j_E(\kappa)$.

- 1 It suffices to argue that $j_U[G(\kappa)] \subseteq H(\kappa_1)$.
- 2 Fix the parameters witnessing the Self-Coding of $\mathbb{Q}_\kappa$: an enumeration $\langle q_\alpha : \alpha < \kappa^{++} \rangle$ of $\mathbb{Q}_\kappa$, and a list $\langle S_\alpha : \alpha < \kappa^{++} \rangle$ of pairwise almost disjoint stationary subsets of $\kappa^+ \cap \text{cf}(< \kappa)$.

Claim

$$H(j_E(\kappa)) = j_U(G)(j_E(\kappa)).$$

Proof Sketch

For simplicity, we denote $\kappa_1 = j_E(\kappa)$.

- 1 It suffices to argue that $j_U[G(\kappa)] \subseteq H(\kappa_1)$.
- 2 Fix the parameters witnessing the Self-Coding of $\mathbb{Q}_\kappa$: an enumeration $\langle q_\alpha : \alpha < \kappa^{++} \rangle$ of $\mathbb{Q}_\kappa$, and a list $\langle S_\alpha : \alpha < \kappa^{++} \rangle$ of pairwise almost disjoint stationary subsets of $\kappa^+ \cap \text{cf}(< \kappa)$.
- 3 Assume that for some $\alpha < \kappa^{++}$, $q_\alpha \in G(\kappa)$, but $j_U(q_\alpha) \notin H(\kappa_1)$.

Claim

$$H(j_E(\kappa)) = j_U(G)(j_E(\kappa)).$$

Proof Sketch

For simplicity, we denote $\kappa_1 = j_E(\kappa)$.

- ① It suffices to argue that $j_U[G(\kappa)] \subseteq H(\kappa_1)$.
- ② Fix the parameters witnessing the Self-Coding of $\mathbb{Q}_\kappa$: an enumeration $\langle q_\alpha : \alpha < \kappa^{++} \rangle$ of $\mathbb{Q}_\kappa$, and a list $\langle S_\alpha : \alpha < \kappa^{++} \rangle$ of pairwise almost disjoint stationary subsets of $\kappa^+ \cap \text{cf}(< \kappa)$.
- ③ Assume that for some $\alpha < \kappa^{++}$, $q_\alpha \in G(\kappa)$, but $j_U(q_\alpha) \notin H(\kappa_1)$.
Then:
 - In $V[G]$, $S_{2\alpha}$ is nonstationary in κ^+ , and $S_{2\alpha+1}$ is stationary in κ^+ .
 - In $M[H]$, $j_U(S_{2\alpha})$ is stationary in κ_1^+ , and $j_U(S_{2\alpha+1})$ is nonstationary in κ_1^+ .

Proof Sketch - Continued

- ④ Therefore, in order to derive a contradiction, we should prove that for every $i < \kappa^{++}$,

Proof Sketch - Continued

- ④ Therefore, in order to derive a contradiction, we should prove that for every $i < \kappa^{++}$,
If S_i is a stationary subset of κ^+ in $V[G]$, then $j_U(S_i)$ is a stationary subset of κ_1^+ in $M[H]$.

Proof Sketch - Continued

- ④ Therefore, in order to derive a contradiction, we should prove that for every $i < \kappa^{++}$,
If S_i is a stationary subset of κ^+ in $V[G]$, then $j_U(S_i)$ is a stationary subset of κ_1^+ in $M[H]$.
- ⑤ By the Kunen-like blueprint, each stationary set S_i concentrate on ordinals of cofinality strictly below κ .

Proof Sketch - Continued

- ④ Therefore, in order to derive a contradiction, we should prove that for every $i < \kappa^{++}$,
If S_i is a stationary subset of κ^+ in $V[G]$, then $j_U(S_i)$ is a stationary subset of κ_1^+ in $M[H]$.
- ⑤ By the Kunen-like blueprint, each stationary set S_i concentrate on ordinals of cofinality strictly below κ .
- ⑥ Given a club $C \subseteq \kappa_1^+$ in $M[H]$, $(j_U)^{-1}[C]$ is $< \kappa$ -club in κ^+ in $V[G]$.

Proof Sketch - Continued

- ④ Therefore, in order to derive a contradiction, we should prove that for every $i < \kappa^{++}$,
If S_i is a stationary subset of κ^+ in $V[G]$, then $j_U(S_i)$ is a stationary subset of κ_1^+ in $M[H]$.
- ⑤ By the Kunen-like blueprint, each stationary set S_i concentrate on ordinals of cofinality strictly below κ .
- ⑥ Given a club $C \subseteq \kappa_1^+$ in $M[H]$, $(j_U)^{-1}[C]$ is $< \kappa$ -club in κ^+ in $V[G]$. It intersects S_i , and applying j_U on the intersection point yields that $C \cap j_U(S_i) \neq \emptyset$.

Open Questions

- Could $\text{UA} + \kappa$ is measurable $+ 2^\kappa = \kappa^{++}$ be achieved from $\mathfrak{o}(\kappa) = \kappa^{++}$?
- Could $\text{UA} + \kappa$ is measurable $+ 2^\kappa = \kappa^{++}$ be forced with a more complex measure structure? (for instance, several normal measures on κ , rather than a unique one).
- Could UA be preserved with $2^\kappa = \kappa^{+3}$ where κ is measurable?

Thank you for your attention!